

Element B

Greenhouse project

1. We have evaluated other products that are similar in some way.

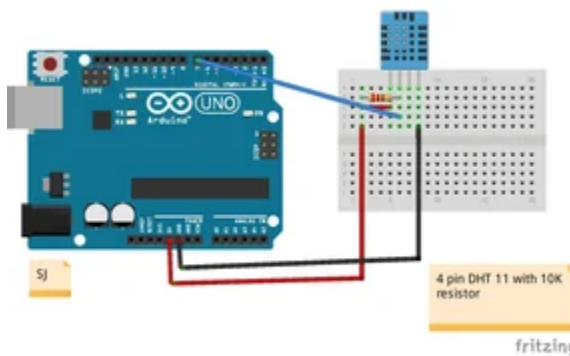
https://www.homedepot.com/p/AcuRite-Digital-Humidity-and-Temperature-Comfort-Monitor-00619HD/202260980?MERCH=REC-_pipsem_-204350179_-0_-n/a_-n/a_-n/a_-n/a_-n/a



humidity: 16% Relative Humidity to 98% relative humidity

temperature: 32°F to 122°F, 0°C to 50°C

It doesn't log temperature and humidity, no data can be stored (only record maximum/minimum in the day).



2.

```
#include <dht.h>
dht DHT;
#define DHT11PIN 7 // sets pin 7 for DHT11 signal connection
void setup() {
  Serial.begin(9600); //opens serial
}
void loop()
{
  int chk = DHT.read11(DHT11PIN); //reads DHT11
```

```
Serial.print(DHT.temperature,0);//prints temp in serial  
Serial.print(",");//prints comma in serial  
Serial.print(DHT.humidity,0);//prints humidity in serial  
Serial.println();//carriage return  
    delay(2000);//wait 2 seconds  
}
```

<https://www.instructables.com/Temperature-and-Humidity-Display-and-Data-Collect-i/>

It successfully recorded temperature and humidity every 2 seconds.

It uses pin 7 for DHT11, and reads the temperature, humidity, then it prints temperature, then comma, and finally humidity.

3. Scholarly Article

https://www.scitechnol.com/peer-review/greenhouse-monitoring-and-control-system-cqrk.php?article_id=20532

The project is to design a sensor-based automated greenhouse conditions control system. They use an Arduino for both sensing and controlling climate conditions including temperature, light intensity, and humidity. The difference between this design and our design was that their design was not meant to be portable, and it would be hooked up to a power supply. This design ultimately succeeded at achieving its goal.